



**United
Nations**

Department of
Economic and
Social Affairs

CLEWs Philippines: updates and way forward

Workshop on Integrated OG-CLEWs Modelling Framework in the Philippines

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Outline

- Model updates
- Scenario implementation: parameters, constraints, and User-Defined Constraints (UDC)
- Data requirements
- Further updates

CLEWs Philippines 2025

- **Original work by (Dineen, C., 2025)**
 - Based on the Model Development Infrastructure (MDI) for CLEWs (Avgerinopoulos et al., 2025)
- **Focus on the energy system representation**
 - Partial implementation of the CLEWs++ approach
 - Sectors represented: power, building, other industry, road transport, other transport (shipping, aviation and rail), agriculture, fuels and heating.
 - The model does not include waste and crop production.
- **Coverage of CLEW systems:**
 - Water use for cooling thermal power plants (m3 per year)
 - Solar PV land footprint (utility scale ground-mounted), using land use intensity coefficient (km2/MW)
 - GHG emissions (MtCO2e per year) with emission factors applied to fuel use or generation. Emissions are not considered for process emissions, land-use change and non-modelled sectors (e.g., crop production).

Global assumptions

- **National level model** (no regional or island disaggregation)
- **Modelling period:** 2020 – 2053
- **Reference year:** 2020
- **Discount rate:** 5%
- **Timeslices** (partition of the year): 30 (monthly and day type divisions)
- **Demands:** base year 2020, data from UN statistics and projected using MDI-derived growth rates and sectoral splits (e.g., electricity, heating and cooking for housing).
- **Techno-economic data** (cost data, efficiencies, etc) from MDI global dataset
- **Emission Factors:**
 - Only in fuel use technologies, CCS and Direct Air Capture (DAC)
 - No emissions from agriculture

Model update (I)

Update	Parameters affected	Data source
Overall revision of the model	Several parameters were updated as data was sometimes missing	-
Calibration of electricity generation by power plant type in the period 2020-2024	Total Technology Annual Activity Upper and Lower Limits	PHL Dept of Energy (DOE) Statistics
Historic capacity and decommissioning schedule of existing power plant	Residual Capacity	PHL DOE (grid-connected installed capacity for Mindanao, Luzon and Visayas)
Capital costs of power sector technologies and transmission and distribution: updated costs considered in the model were generic and sometimes underestimated	Capital Cost	Lazard, NREL Annual Technology Cost Database 2021, IRENA

Model updates (II)

Update	Parameters affected	Data source
In some cases operational life was missing, or the default value of 25 was considered. The update was implemented in power sector technologies.	Operational Life	Lazard, NREL Annual Technology Cost Database 2021, and assumption based on current operation of power plants vs their start year (E.g., coal power plant – 40years)
Revision and update of CO ₂ -eq emission factors (sometimes missing or allocated multiple times)	Emission Factor, emission: CO ₂ e	IPCC GHG Inventory Guidelines
Creation of PM _{2.5} emissions and assignment to power, transport and cookstoves technologies using fossil fuels	Emission Factor for the Emission PM _{2.5}	Emission factors based on EMEP/European Environment Agency EMEP/EEA air pollutant emission inventory guidebook 2023

Scenario implementation

Scenario	Description
Baseline (Base)	No policy constraints. Projections according to MDI tools
Philippines Energy Plan (PEP)	RE targets: 35% by 2030, and 50% by 2040, with no less than 50% onward Coal phase-out: Decommissioning of existing coal generation capacity by 2040 Fleet electrification: 50% by 2040 onwards

CLEWs Philippines 2025: Scenarios

Scenario	Description
Baseline (Base)	No policy constraints. Projections according to MDI tools For the representation of RE potential, electricity generation limits were implemented in the Total Annual Upper Limit to ensure investments did not exceed a feasible potential.

CLEWs Philippines 2025: Scenarios

Scenario	Description
Philippines Energy Plan (PEP)	RE targets: 35% by 2030, and 50% by 2040, with no less than 50% onward Coal phase-out: Decommissioning of existing coal generation capacity by 2040 Fleet electrification: 50% by 2040 onwards

RE targets introduced using a User-Defined Constraint (which is created in “model configuration in MUIO”, and enables creating constraints beyond the ones existing in the OSeMOSYS code.

In this case, the user-defined constraint identified the technologies to which the constraint was applied, and defined the % of each technology that would contribute to electricity generation each year, considering the targets in the policy (see table above).

CLEWs Philippines 2025: Scenarios

Scenario	Description
Philippines Energy Plan (PEP)	RE targets: 35% by 2030, and 50% by 2040, with no less than 50% onward Coal phase-out: Decommissioning of existing coal generation capacity by 2040 Fleet electrification: 50% by 2040 onwards

- **Coal phase-out of existing coal power plants** was created using the last year coal power plant generation (from 2024) and reducing it linearly to 0 by 2040.
- This was implemented in the parameter “Total Technology Annual Upper Limit” for the technology representing old coal power plants.

CLEWs Philippines 2025: Scenarios

Scenario	Description
Philippines Energy Plan (PEP)	RE targets: 35% by 2030, and 50% by 2040, with no less than 50% onward Coal phase-out: Decommissioning of existing coal generation capacity by 2040 Fleet electrification: 50% by 2040 onwards

- For **Fleet electrification**, also User-Defined Constraints were used, which affected all road transport technologies that used electricity.
- Road transport “Accumulated Annual Demands” (an input to the model) were used to estimate the annual values of transport demand to be met by electric vehicle technologies according to the 50% electrification target by 2040, starting from 0 in 2026.
- A 50% electrification was assumed for the remainder period (2040.2050)



CLEWS+ + Represents all Major Conventional, Low-Carbon and Zero Emissions Technologies to meet demand in Future Energy Scenarios

Power

- **Generation** Coal | Oil | Gas | Biomass | Coal with CCS | Gas with CCS | Biomass with CCS | Nuclear | Hydro | Onshore wind | Offshore wind | Solar | Hydrogen
- **Storage** Battery storage | Power to gas

Buildings

- **Residential heat | Non-residential heat** Coal boiler | Oil boiler | Gas boiler | Biomass boiler | District heat | Resistive heater | Heat pump
- **Residential cooking** Biomass stove | Coal stove | Gas stove | Oil stove | Electric stove
- **Residential appliances | Non-residential appliances**

Transport

- **2-3 wheelers | Cars | Buses | Light commercial vehicles | Heavy commercial vehicles** Liquid fuel ICE | Plug-in hybrid | Battery electric | Hydrogen fuel cell | Natural gas ICE
- **Rail** Liquid fuel | Electric
- **Aviation** Liquid fuel
- **Shipping** Liquid fuel | Ammonia

Liquid fuel production

- Petroleum-derived fuel | Biofuel | Synthetic fuel

Industry

- **Iron and steel** Blast furnace-basic oxygen furnace (BF-BOF) | BF-BOF with CCS | Gas direct reduced iron with electric arc furnace (DRI-EAF) | Hydrogen DRI-EAF | Scrap steel with electric arc furnace
- **Cement kilns** Coal | Oil | Gas | Biomass | Coal with CCS | Gas with CCS | Biomass with CCS | Hydrogen
- **Low-temp process heat** Coal | Oil | Gas | Biomass | Hydrogen | Heat pump | District heat
- **High-temp process heat** Coal | Oil | Gas | Biomass | Coal with CCS | Gas with CCS | Biomass with CCS | Hydrogen
- **High value chemicals** Oil

Hydrogen

- Coal gasification | Steam methane reformer | Coal gasification with CCS | Steam methane reformer with CCS | Electrolyzer

GHG removal

- Direct air capture

Other sectors

- **Agriculture** Soils | Enteric fermentation | Manure management
- **Land use change**
- **Waste** Solid waste | Liquid waste



List of Major Data Flows in CLEWs

Input

Output

Costs		• Technology costs • Fuel costs	• Investment needs by year • Running costs by year
Energy		• Energy demand (often service demand) • Initial conditions (calibration) • Energy intensities/efficiencies	• Total demand by fuel and by sector • Total energy intensity by sector
Emissions		• Emission intensities • Emission limits • Emission prices	• Emissions by sector/technology • Emission intensity by sector
Land use		• Total land area and initial breakdown • Crop demand	• Land use change • Crop production
Waste		• Waste production and initial treatment method breakdown	• Future waste treatment method breakdown

Data requirements

Energy	Water	Land	Climate
Final energy consumption by sector (demands)	Water consumption in the industry	Forestry representation	CO2eq emission factors (implemented in fossil fuel sectoral demand technologies)
Power plants technoeconomic data	Water consumption in the power sector	Crop yields	CO2 removals
Fuel prices	Water sectoral demands (potable)	Land cover	Precipitation
Renewable energy potential	Energy requirements for water systems	Water balance outputs per land cover type	Other emission factors
Electricity flows between main island groups	Technoeconomic data for water treatment and supply technologies	Crop cultivation costs	
New infrastructure	Potable water sectoral tariffs		
Time representation and load profile	Water use by source		

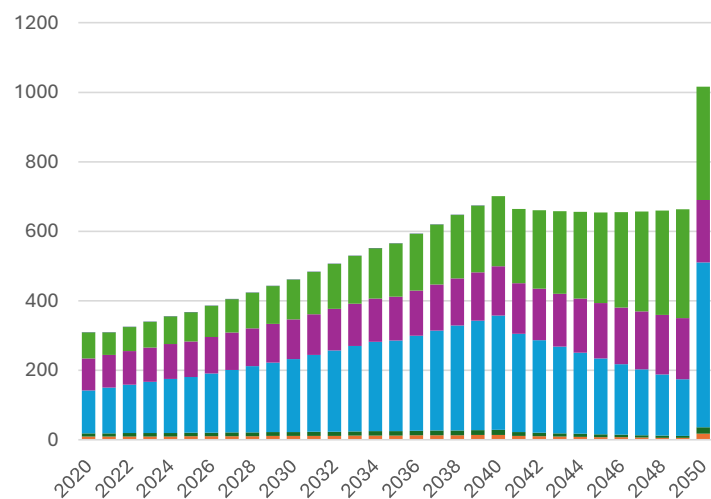
Further updates

- Additional calibration of the first 5 years of model (2020-2025) for all energy uses across sectors represented.
- Inventory of process efficiencies across sectors represented, and update of Input Activity Ratios (i.e., how much of a commodity is used in a process)
- Exogenous demands revision and recalculation of demand projections
- Revision of Baseline and PEP scenario assumptions of how sectors evolve over the modelling period
- Expand the model coverage:
 - Complete representation of sectors considered in the NDCs, as respective emissions were missing (Waste and agriculture)
 - Representation of crop cultivation and other land use-related sector
- Update overall model data using official data sources
- Update the PEP scenario (no coal power plant investments beyond 2030, EV user-defined constraints created for each transport category (e.g., trucks, buses, cars, etc))
- In the future, possibly consider representing regions of the Philippines, instead of an aggregated national model

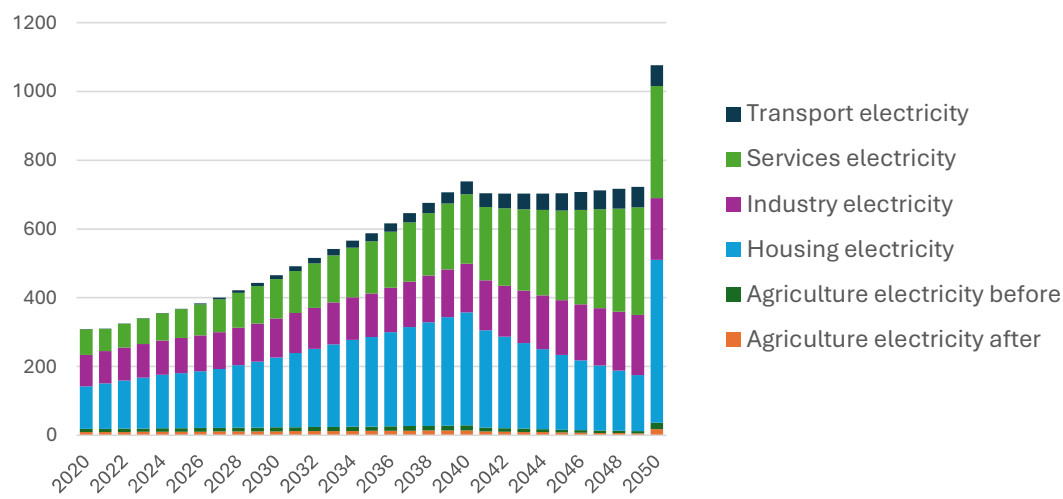


Electricity demands (model version v8)

Electricity demand - Base scenario (PJ)



Electricity demand - PEP scenario (PJ)

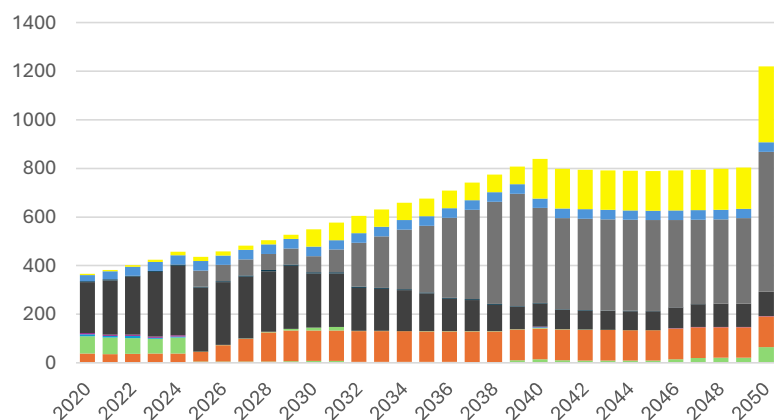


- Includes endogenous and exogenous demands.
- PEP scenario – additional demand for transport.

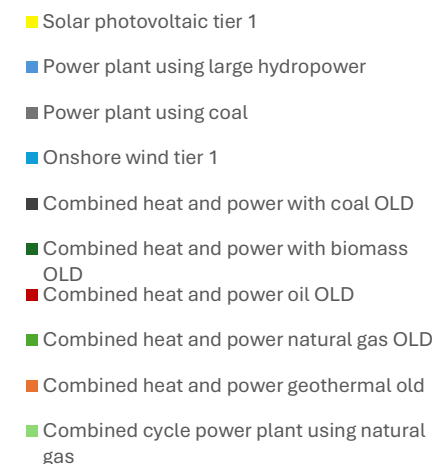
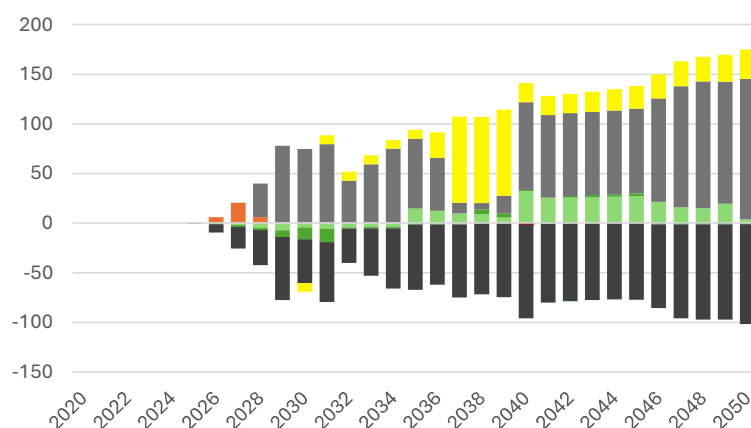


Scenario comparison: Electricity generation (model version v8)

Electricity generation - Base (PJ)



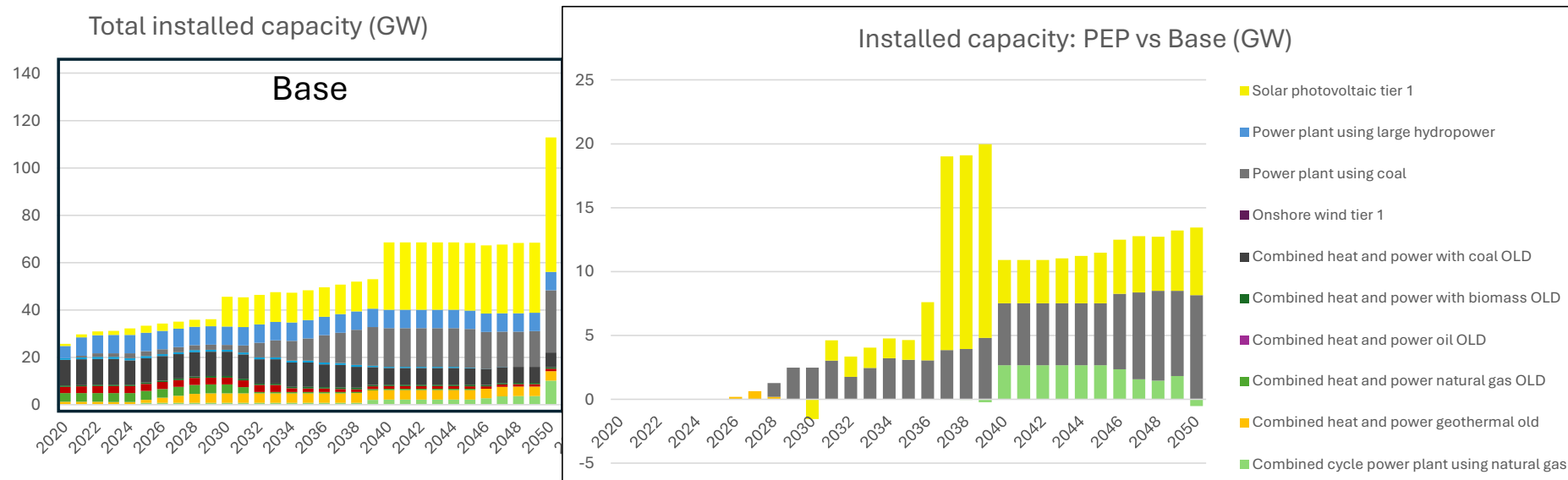
Electricity generation - PEP vs Base (PEP) (PJ)



- Difference in generation due to transport electricity demand (negligible) and phase-out of old coal plants.
- Installation of new coal plants, solar and natural gas



Scenario comparison: Installed power capacity (model version v8)

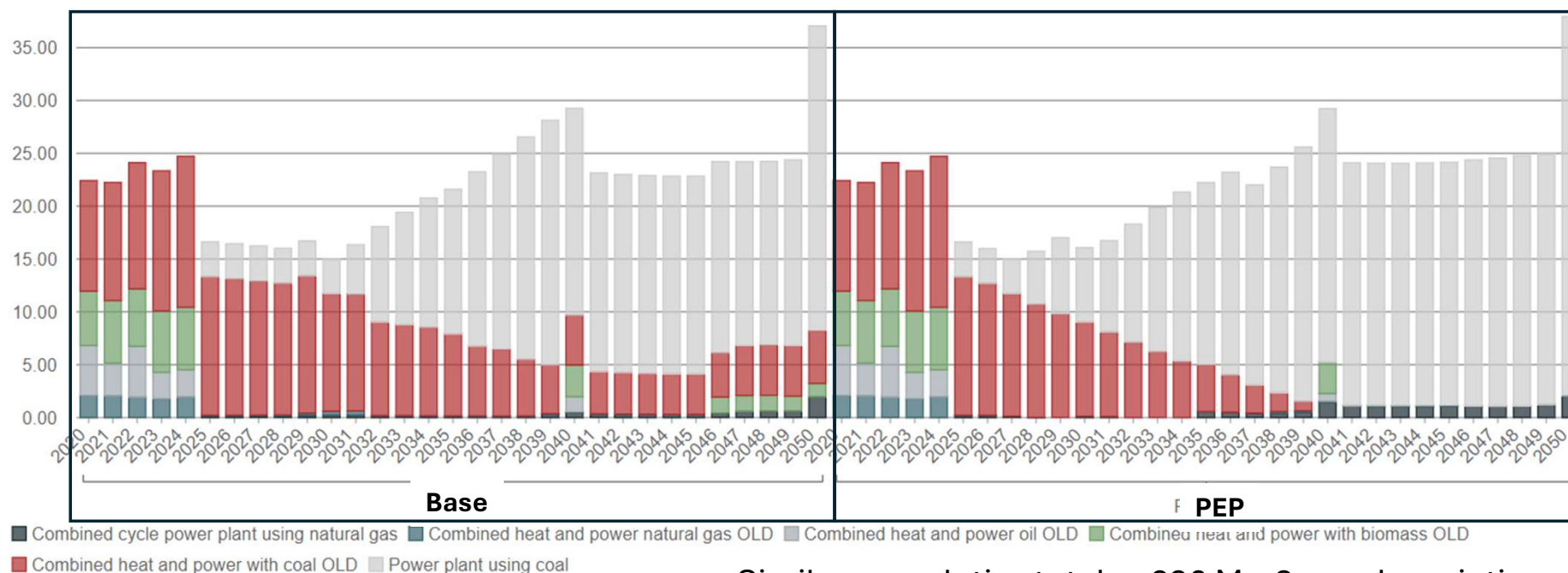


- Coal and solar PV represent the technologies with the largest installed capacity
- Similar trend in PEP, with higher additional capacity from these technologies, plus natural gas
- Assess the feasibility of capacity installation/year



Scenario comparison: Water use by thermal power plants (model version v8)

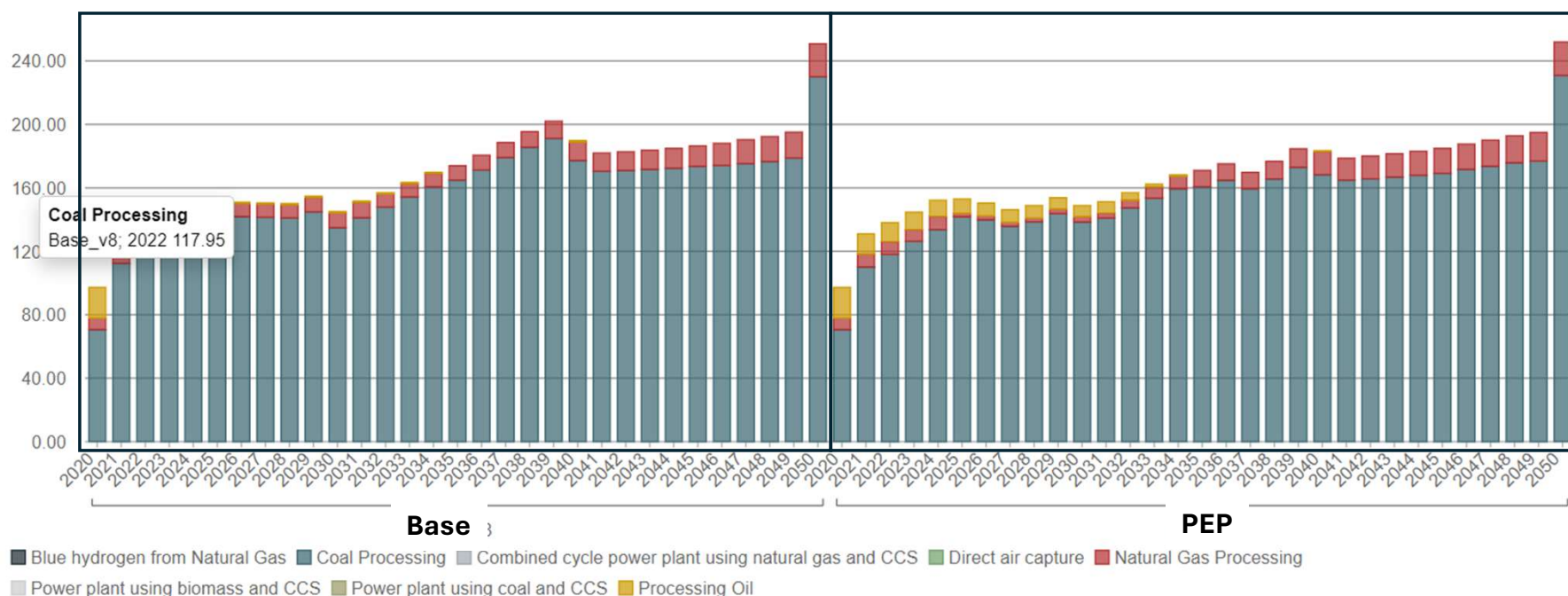
Water use for cooling in power plants (Mm3)



- Similar cumulative totals ~690 Mm3, yearly variations
- Water use dominated by coal power plant operation



Scenario comparison: CO₂-eq emissions (model version v8)



- Most of the emissions from the power sector (coal power plants)
- 5,273 Mton in the Base vs 5,188 Mton in PEP (2% reduction)



Thank you